

Shubham Sarkar

Data Scientist

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SUMMARY

Data Scientist with experience designing and deploying end to end software with frontend ,backend, database and AI functionality. Strong background in Machine Learning, statistics, Deep Learning ,Natural Language Processing, Computer Vision and Deployment. Ability to process and analyze unstructured data, build and fine tune agents, and to deliver production-ready AI systems.

SKILLS

Machine Learning & AI: Supervised & Unsupervised Learning, NLP, Computer Vision, LLM Fine Tuning, Retrieval Augmented Generation(RAG)

Frameworks & Libraries: TensorFlow, PyTorch, scikit-learn, Hugging Face, Pandas, NumPy,Langchain

MLOps & Deployment:Amazon Web Services (AWS), Docker, MLflow, DVC, FastAPI, Flask, Streamlit, CI/CD

Data & Systems: SQL, ETL Pipelines, Linux, AWS, Vector Databases

Programming : Python,C, Javascript,HTML,CSS,Data Structures and Algorithms

WORK EXPERIENCE

Machine Learning Engineer , Alignerr

Jan 2026–Present

- Enhanced model prediction accuracy by 15% by analyzing agent responses on different rubrics and state scenarios.
- Transcribed 50+ audio and video files to provide quality training and test data.
- Evaluated LLM conversations to assess correctness of tool usage, API calls, and reasoning flow

PROJECTS

Tailorcv.ai

Github:github.com/SSarkar-1/TailorCV.ai , Website: <https://tailorcv-ai.onrender.com>

- Built an AI web app that tailors resumes to job descriptions using LLMs and NLP pipelines, improving resume relevance by up to 80%.
- Designed a Python and FastAPI backend and HTML CSS and Javascript for frontend and deployed on AWS
- Got 50+ users within the first week of launch, demonstrating strong early adoption and real-world impact.

Youtube Sentiment Analysis

Github:https://github.com/SSarkar-1/Youtube_Sentiment_Insights, Demo: [Demo](#)

- Created an end-to-end YouTube sentiment analysis pipeline processing 10k+ user comments, improving sentiment classification performance through NLP preprocessing (tokenization, lemmatization, stopword removal).
- Tracked multiple model experiments using MLflow and DVC, enabling reproducible training and systematic comparison of models built with scikit-learn and NLP libraries.
- Deployed the pipeline on AWS using Docker and exposed predictions via Flask REST APIs, enabling scalable and reproducible inference.

RAG System

Github:github.com/SSarkar-1/RAG

- Made a production-ready RAG pipeline integrating semantic vector retrieval with LLM generation to produce context-grounded responses.
- Engineered multiple chunking strategies and a scalable ingestion , retrieval , generation flow for efficient semantic search and generation.
- Implemented history-aware and multimodal augmentations, and evaluated retrieval outputs to measure relevance and quality.

Smart Product Pricing Model, Amazon ML Challenge 2025

Kaggle : <https://www.kaggle.com/code/shubhamsarkarss/multimodal-working>

- Created a NLP and CV pipeline to analyze 150,000 image and text data using transformer-based text encoders and CNN-based image embeddings, integrating them through a fusion neural network for price prediction
- Implemented data preprocessing, including text cleaning, tokenization, and streaming image feature extraction with ResNet and CLIP representations to handle large dataset.
- Applied feature engineering, outlier handling, and SMAPE-based evaluation to optimize prediction accuracy.
- Built and fine-tuned models using TensorFlow, and scikit-learn, accomplished 142 rank. out of 50,000 participants.

EDUCATION

Bachelor of Technology ,Jadavpur University

2023–2027

Secured 98.5 percentile in jee mains out of 1,400,000 students and 1458 rank in WBJEE

High School , Hariyana Vidya Mandir

2020–2022

Scored 96% in class 10 to get into one of the best schools in Kolkata. 92% in class 12.

EXTRACURRICULARS

Core Member, Jadavpur University Entrepreneurship Cell

•Organized national-level flagship events such as E-Summit 2025 and Hult Prize 2025, achieving 5,000+ registrations and 1,000+ on-campus attendees.

•Contributed to the establishment of an Incubation Center at Jadavpur University under the Institution’s Innovation Council (IIC)
Coordinator, Jadavpur University Finance Club

•Led planning and execution of Finspire 2025, a national-level finance event with 1,000+ registrations and 500+ on-ground attendees, strengthening the club’s national presence.

•Delivered high-impact trading and investment courses to 100+ students, improving engagement in financial markets